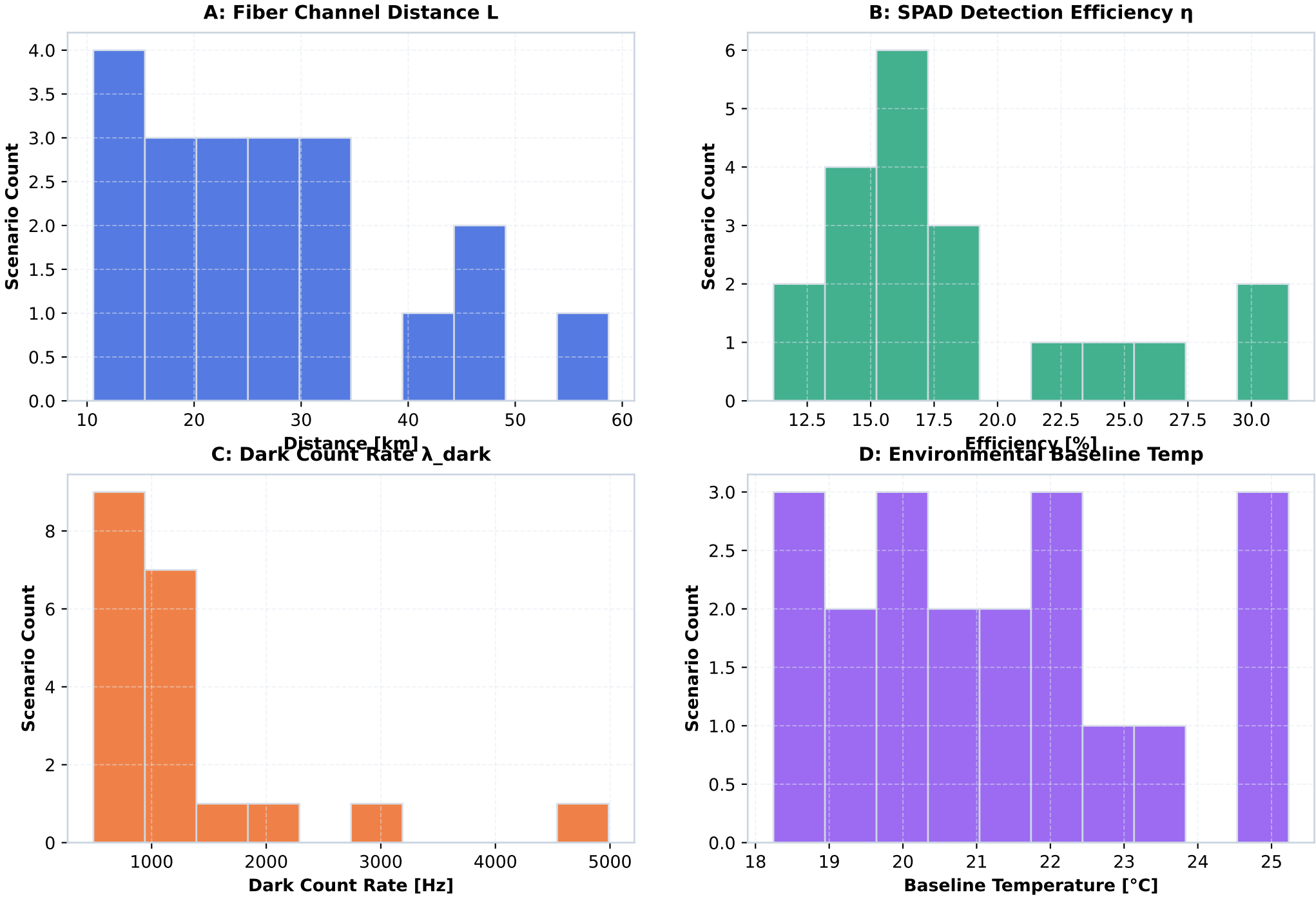Population Diversity: Latin Hypercube Sampling (LHS) Parameter Coverage Across Scenarios